Importing Necessary Libraries

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
plt.style.use('Solarize_Light2')

%matplotlib inline
```

Loading the Dataset

```
df = pd.read_csv(r'C:\Users\yasha\OneDrive\Desktop\inlighntech.com\My projects\Python\
Zomato\Zomato data.csv')
df.head()
```

```
                    name online_order book_table   rate  votes  \
0                  Jalsa          Yes        Yes  4.1/5    775
1         Spice Elephant          Yes         No  4.1/5    787
2        San Churro Cafe          Yes         No  3.8/5    918
3  Addhuri Udupi Bhojana           No         No  3.7/5     88
4          Grand Village           No         No  3.8/5    166

   approx_cost(for two people) listed_in(type)
0                          800          Buffet
1                          800          Buffet
2                          800          Buffet
3                          300          Buffet
4                          600          Buffet
```

```
df.shape
```

```
(148, 7)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148 entries, 0 to 147
Data columns (total 7 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   name                         148 non-null    object
 1   online_order                 148 non-null    object
 2   book_table                   148 non-null    object
 3   rate                         148 non-null    object
 4   votes                        148 non-null    int64
 5   approx_cost(for two people)  148 non-null    int64
 6   listed_in(type)              148 non-null    object
dtypes: int64(2), object(5)
memory usage: 8.2+ KB
```

```
df.describe()
```

```
            votes  approx_cost(for two people)
count  148.000000                   148.000000
mean   264.810811                   418.243243
```

```
std     653.676951                   223.085098
min       0.000000                   100.000000
25%       6.750000                   200.000000
50%      43.500000                   400.000000
75%     221.750000                   600.000000
max    4884.000000                   950.000000
```

Data Cleaning and preparation

Convert Ratings:

```
#checking rate column
df['rate'].unique()

array(['4.1/5', '3.8/5', '3.7/5', '3.6/5', '4.6/5', '4.0/5', '4.2/5',
       '3.9/5', '3.1/5', '3.0/5', '3.2/5', '3.3/5', '2.8/5', '4.4/5',
       '4.3/5', '2.9/5', '3.5/5', '2.6/5', '3.8 /5', '3.4/5'],
      dtype=object)

# Splitting and converting rate column to numeric format
df['rate'] = df['rate'].str.split('/').str[0]
df['rate'] = df['rate'].astype(float)

# checking again after conversion
df['rate'].unique()

array([4.1, 3.8, 3.7, 3.6, 4.6, 4. , 4.2, 3.9, 3.1, 3. , 3.2, 3.3, 2.8,
       4.4, 4.3, 2.9, 3.5, 2.6, 3.4])
```

Verify Data Types:

```
# checking the Datatypes of all columns
df.dtypes

name                            object
online_order                    object
book_table                      object
rate                           float64
votes                            int64
approx_cost(for two people)      int64
listed_in(type)                 object
dtype: object

# Unique Values in all columns
for col in df.columns:
        print(f"\nUnique values in {col}:")
        print(df[col].unique())


Unique values in name:
['Jalsa' 'Spice Elephant' 'San Churro Cafe' 'Addhuri Udupi Bhojana'
 'Grand Village' 'Timepass Dinner'
 'Rosewood International Hotel - Bar & Restaurant' 'Onesta'
 'Penthouse Cafe' 'Smacznego' 'Village Café' 'Cafe Shuffle'
 'The Coffee Shack' 'Caf-Eleven' 'Cafe Vivacity' 'Catch-up-ino'
 "Kirthi's Biryani" 'T3H Cafe' '360 Atoms Restaurant And Cafe'
 'The Vintage Cafe' 'Woodee Pizza' 'Cafe Coffee Day' 'My Tea House'
```

```
 'Hide Out Cafe' 'CAFE NOVA' 'Coffee Tindi' 'Sea Green Cafe' 'Cuppa'
 "Srinathji's Cafe" 'Redberrys' 'Foodiction' 'Sweet Truth'
 'Ovenstory Pizza' 'Faasos' 'Behrouz Biryani' 'Fast And Fresh'
 'Szechuan Dragon' 'Empire Restaurant' 'Maruthi Davangere Benne Dosa'
 'Chaatimes' 'Havyaka Mess' "McDonald's" "Domino's Pizza" 'Hotboxit'
 'Kitchen Garden' 'Recipe' 'Beijing Bites' 'Tasty Bytes' 'Petoo'
 'Shree Cool Point' 'Corner House Ice Cream' 'Biryanis And More'
 'Roving Feast' 'FreshMenu' 'Banashankari Donne Biriyani' 'Wamama'
 'Five Star Chicken' 'XO Belgian Waffle' 'Peppy Peppers' 'Goa 0 Km'
 'Chinese Kitchen' 'Jeet Restaurant' 'Cake of the Day' 'Kabab Magic'
 "Namma Brahmin's Idli" 'Gustoes Beer House' 'Sugar Rush' 'Burger King'
 'The Good Bowl' 'The Biryani Cafe' 'Spicy Tandoor' 'LSD Cafe'
 'Rolls On Wheels' 'Om Sri Vinayaka Chats'
 'Sri Guru Kottureshwara Davangere Benne Dosa'
 'Devanna Dum Biriyani Centre' 'Kolbeh' 'Upahar Sagar'
 'Kadalu Sea Food Restaurant' 'Frozen Bottle' 'Parimala Sweets'
 'Vaishali Deluxe' 'Chill Out' 'The Big O Bakes' 'Meghana Foods'
 'Krishna Sagar' 'Dessert Rose' 'Chickpet Donne Biryani House'
 'Me And My Cake' 'Sunsadm' 'Annapooraneshwari Mess'
 "Thanco's Natural Ice Creams" 'Nandhini Deluxe'
 "Vi Ra's Bar and Restaurant" 'Kaggis' 'Ayda Persian Kitchen'
 'Chatar Patar' 'Polar Bear' "Kidambi's Kitchen" 'Mane Thindi'
 'Kotian Karavali Restaurant' 'Floured-Baked With Love' 'Cakes & Slices'
 'Spice 9' 'Coffee Shopee' 'Naveen Kabab & Biriyani Mane'
 'Katriguppe Donne Biryani' 'Hari Super Sandwich'
 'Atithi Point Ande Ka Funda' 'Just Bake'
 'Dharwad Line Bazaar Mishra Pedha' 'Cake Bite' "Aarush's Food Plaza"
 'Wood Stove' 'Kulfi & More' 'Kannadigas Karavali' 'K27 - The Pub'
 'Bengaluru Coffee House' 'New Mangalore Lunch Home' 'Coffee Bytes'
 'Parjanya Chat Zone' "Kwality Wall's Swirl's Happiness Station"
 'Soms Kitchen & Bakes' 'Banashankari Nati Style' 'Ruchi Maayaka'
 'Mohitesh Hut Roll' 'Sri Basaveshwar Jolada Rotti Oota'
 'Roll Magic Fast Food' 'Foodlieious Multi Cuisine'
 'Thanishka Nati And Karavali Style' 'Swathi Cool Point' 'Kaumudis Juoice'
 'Amma - Manae' 'Sri Sai Tiffannies' 'Hotel Andhra Speices'
 'Sri Murari Family Restaurant' 'Aramane Donne Biriyani' 'Darkolates'
 'Swaada Healthy Kitchen' 'Gawdaru Mane Beriyani' 'Melting Melodies'
 'New Indraprasta' 'Anna Kuteera' 'Darbar' 'Vijayalakshmi']

Unique values in online_order:
['Yes' 'No']

Unique values in book_table:
['Yes' 'No']

Unique values in rate:
[4.1 3.8 3.7 3.6 4.6 4.  4.2 3.9 3.1 3.  3.2 3.3 2.8 4.4 4.3 2.9 3.5 2.6
 3.4]

Unique values in votes:
[ 775  787  918   88  166  286    8 2556  324  504  402  150  164  424
   90  133  144   93   13   62  180   28   31   11   75    4   23  148
  219  506   35  172  415  230   91 1647 4884   17  540   36  244  804
  679  245   21  345  618 1047  627  104  354   55  163   58  808   78
 1720   34  868   39   71    6  520    0   84  299  558   22  153  146
   14   42   66 4401    7    9  283   64   65   52  130   10   45   29
   30  201   25  771   98   47]
```

```
Unique values in approx_cost(for two people):
[800 300 600 700 550 500 450 650 400 900 200 750 150 850 100 350 250 950]

Unique values in listed_in(type):
['Buffet' 'Cafes' 'other' 'Dining']
```

```
# Checking for null values
df.isnull().sum()
```

```
name                           0
online_order                   0
book_table                     0
rate                           0
votes                          0
approx_cost(for two people)    0
listed_in(type)                0
dtype: int64
```

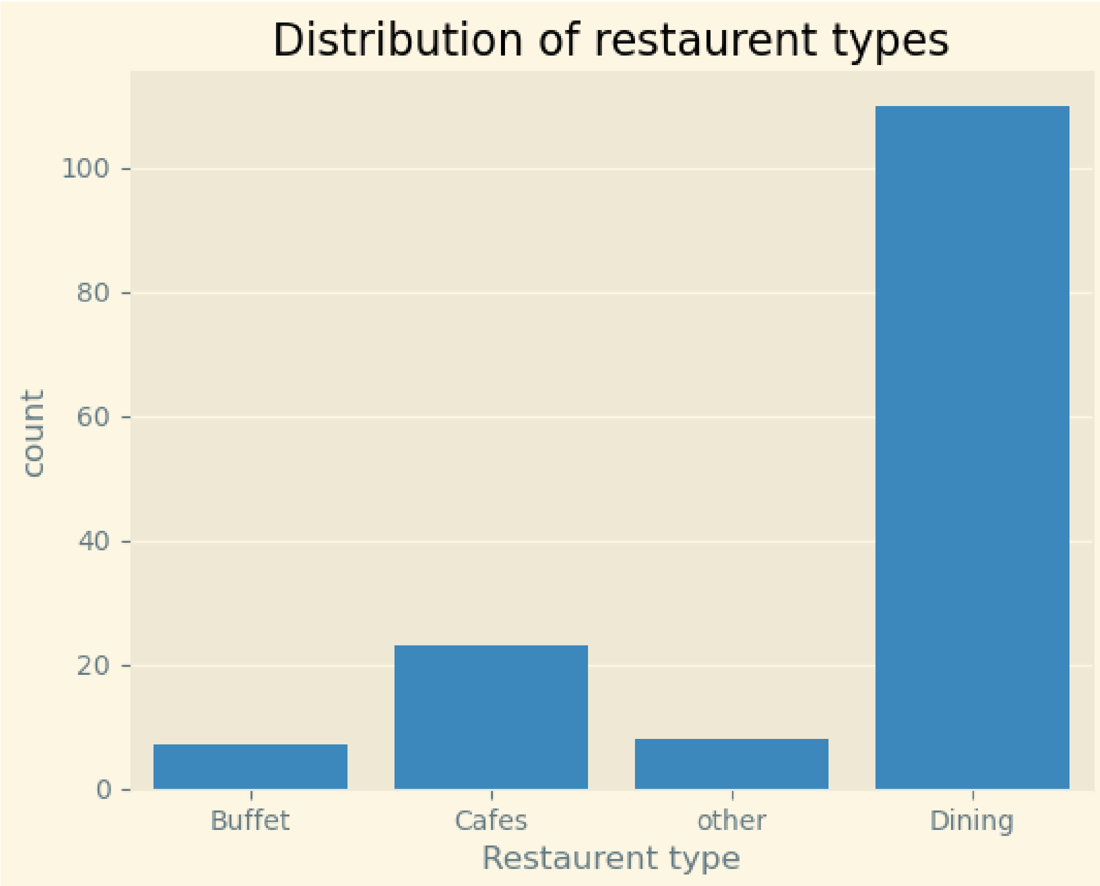
Conclusion:

- Converted the Rate column into float for better analysis
- Verified the datatypes in all columns
- Checked unique values for all columns
- Checked for null values - found and confirmed no null values

Data Analysis and Visualization

Types of Restaurants

```
# count plot for restaurent types
sns.countplot(data=df, x='listed_in(type)')
plt.title('Distribution of restaurent types')
plt.xlabel('Restaurent type')
plt.show()
```



Conclusion:

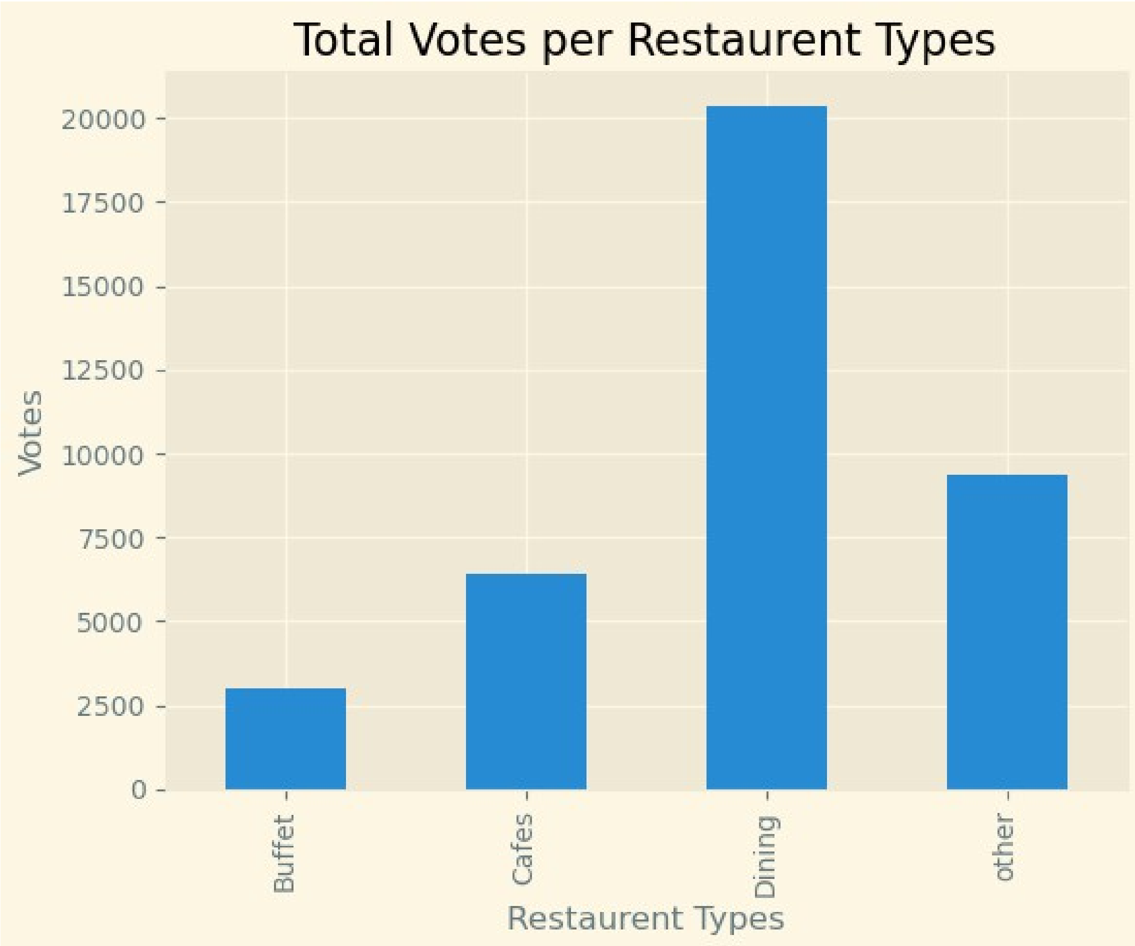
- Dining seems to be most common and more preferred
- Dining contributes to approx 90% of Restaurent distribution
- This strongly suggests that customers prefer sit-down dining experiences over other formats
- Dining establishments are simply far more prevalent in the region
- Cafes are next most preferred restaurant type
- Buffet and other are least preferred

Votes by Restaurant Type

```
votes_by_type = df.groupby('listed_in(type)')['votes'].sum()
votes_by_type
```

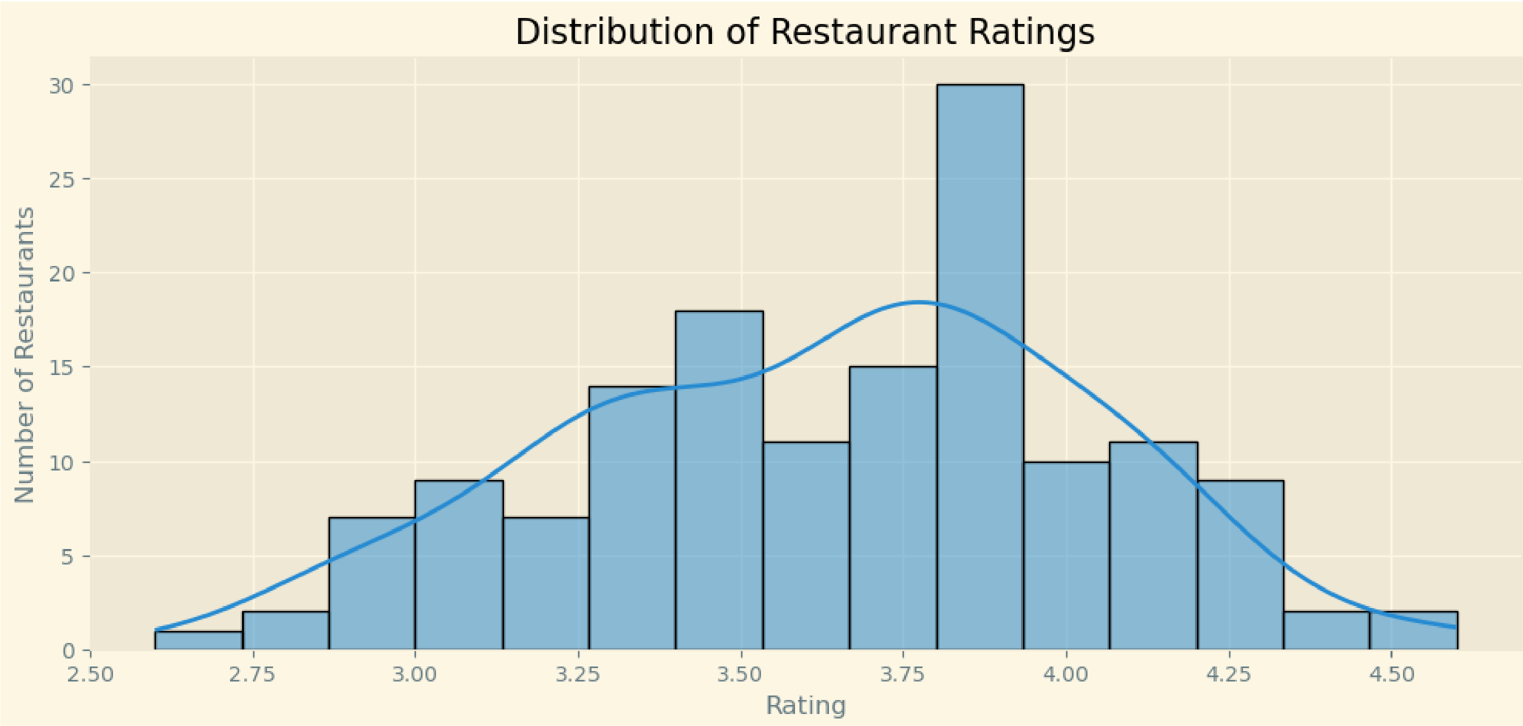
```
listed_in(type)
Buffet      3028
Cafes       6434
Dining     20363
other       9367
Name: votes, dtype: int64
```

```
# Barplot of Total Votes per Restaurent Types
votes_by_type.plot(kind='bar')
plt.title('Total Votes per Restaurent Types')
plt.xlabel('Restaurent Types')
plt.ylabel('Votes')
plt.show()
```



Ratings Distribution

```
# histogram of the rate column
plt.figure(figsize=(12,5))
sns.histplot(data=df, x='rate', kde=True, bins=15)
plt.title("Distribution of Restaurant Ratings")
plt.xlabel("Rating")
plt.ylabel("Number of Restaurants")
plt.show()
```

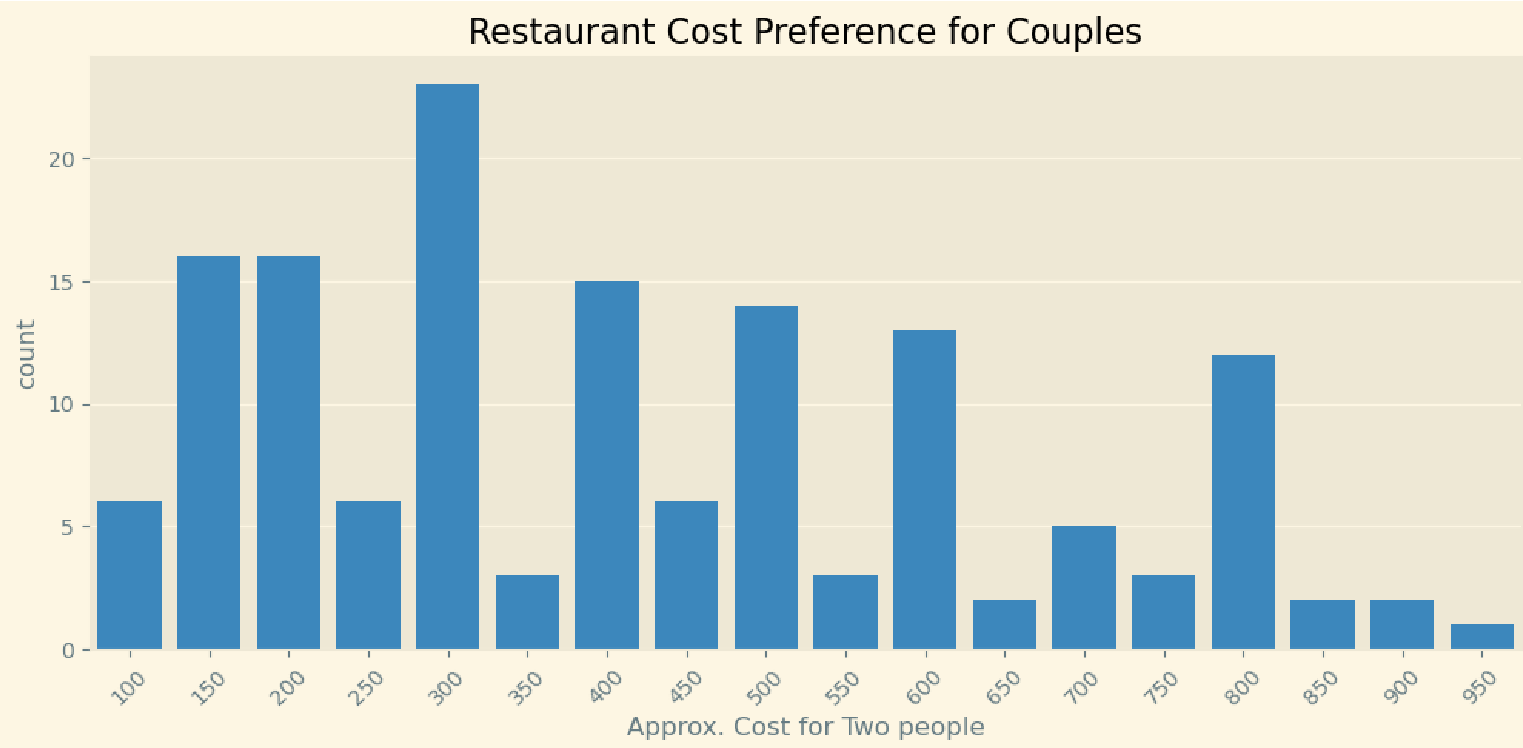


Conclusion:

- The rating range most restaurants fall into is (typically 3.5 to 4)
- This suggests that a rating of 3.5–4.0 is the norm
- The highest rating range for restuarants is 3.8 to 3.95
- Anything above 4.2 would be considered genuinely outstanding

Restaurant Cost Preference for Couples

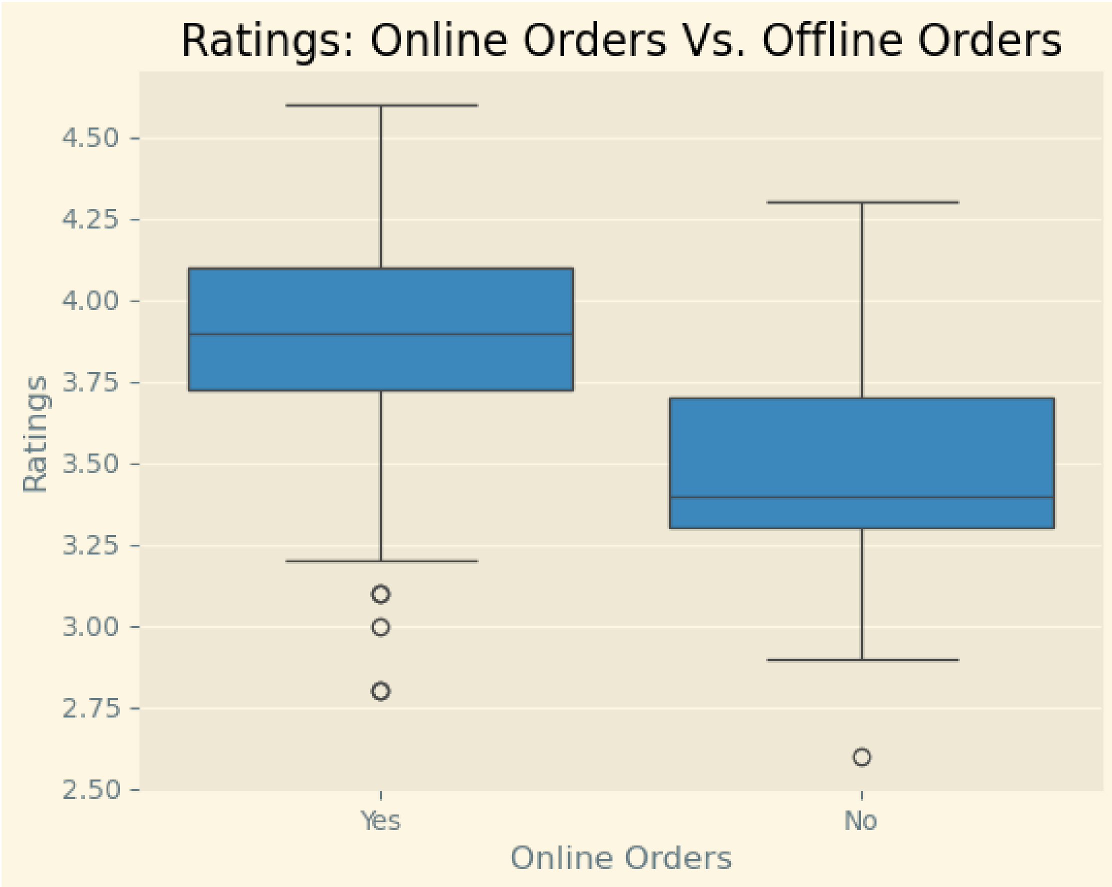
```
plt.figure(figsize=(12,5))
sns.countplot(data=df, x='approx_cost(for two people)')
plt.title('Restaurant Cost Preference for Couples')
plt.xticks(rotation=45)
plt.xlabel('Approx. Cost for Two people')
plt.show()
```



Online vs. Offline Ratings

```
# Boxplot of Ratings: Online Orders Vs. Offline Orders

sns.boxplot(data=df, x='online_order', y='rate')
plt.title('Ratings: Online Orders Vs. Offline Orders')
plt.xlabel('Online Orders')
plt.ylabel('Ratings')
plt.show()
```



Conclusion:

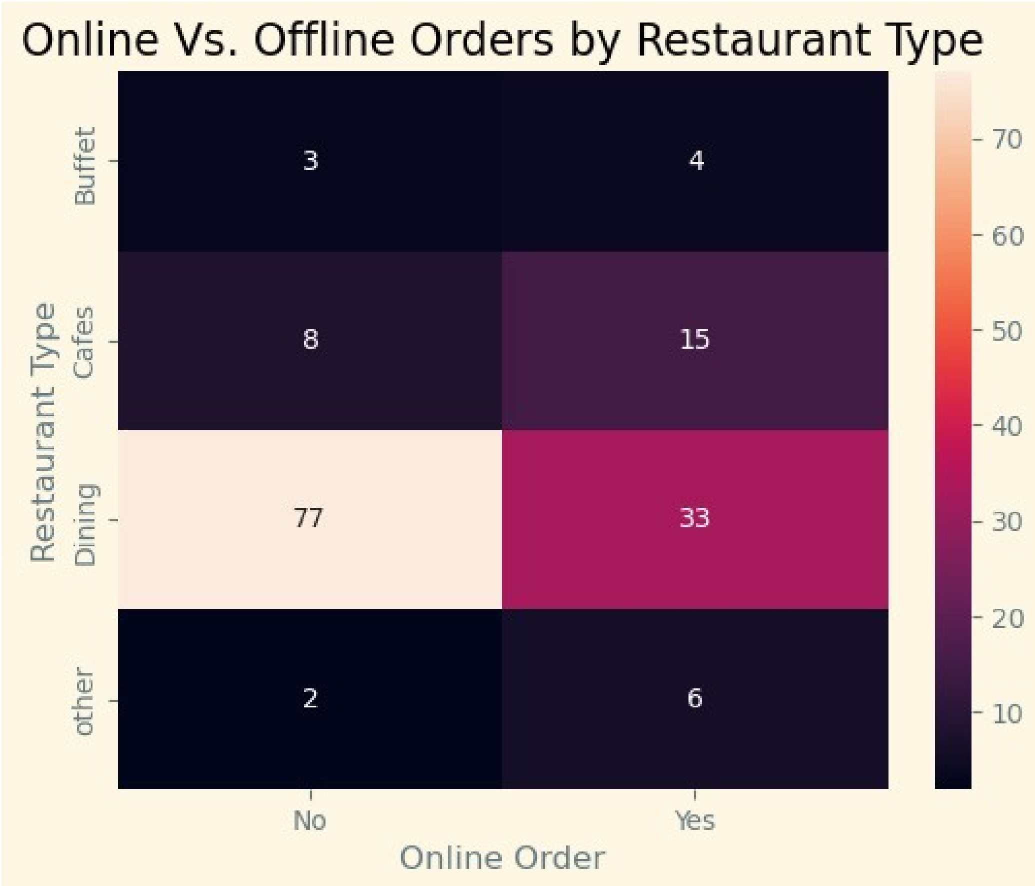
- Online orders have higher ratings than offline ones
- Restaurants that accept online orders tend to receive higher customer ratings compared to those that only serve offline
- This could suggest that the convenience of online ordering contributes to a better overall customer experience
- Restaurants with online ordering tend to be more established/quality-focused

Online Orders by Restaurant Type

```
# Pivot table for No. of restaurant types offering online vs offline orders
pivot = pd.pivot_table(df, index='listed_in(type)', columns='online_order',
values='rate', aggfunc='count')
pivot
```

online_order	No	Yes
listed_in(type)		
Buffet	3	4
Cafes	8	15
Dining	77	33
other	2	6

```
sns.heatmap(data= pivot, annot=True)
plt.title("Online Vs. Offline Orders by Restaurant Type")
plt.xlabel("Online Order")
plt.ylabel("Restaurant Type")
plt.show()
```



Conclusion

- Cafes prefer online orders (15 vs 8 — nearly 2x more online), while Dining restaurants lean heavily toward offline orders
- Likely because cafe customers often pre-order or use delivery apps for convenience
- The brightest cell (Dining–Offline = 77) dominates the heatmap (77 vs 33 — more than 2x more offline)
- Buffet & Other categories have very small counts, but both marginally lean online
- This pattern suggests that restaurant type strongly influences ordering behavior
- Delivery/online platforms are more relevant for cafes, while dining establishments rely on walk-in customers

Find the Insights :

What type of restaurant do the majority of customers order from?

```
majority_type = (df[df['online_order'] == 'Yes']['listed_in(type)'].value_counts())

print(majority_type)
```

```
listed_in(type)
Dining    33
Cafes     15
other      6
Buffet     4
Name: count, dtype: int64
```

- The majority of customers order from Dining restaurants.
- This is evident from the highest count of dining-type restaurants and their strong customer engagement compared to other types such as cafés or delivery-only outlets

How many votes has each type of restaurant received from customers?

```
votes_by_type = (df.groupby('listed_in(type)')
['votes'].sum().sort_values(ascending=False))

votes_by_type
```

```
listed_in(type)
Dining    20363
other      9367
Cafes      6434
Buffet     3028
Name: votes, dtype: int64
```

- Dining restaurants have received the highest total number of votes
- Followed by cafés and other delivery restaurants
- This indicates that dining restaurants attract more customer attention and feedback, reflecting higher popularity

What are the ratings that the majority of restaurants have received?

```
df['rate'].value_counts().sort_values(ascending=False).nlargest(5)
```

```
rate
3.8    20
3.7    15
3.3    14
3.4    12
4.1    11
Name: count, dtype: int64
```

- Most restaurants have received ratings in the range of 3.5 to 4.0
- This suggests that the overall quality of restaurants listed on Zomato is good
- Customers are generally satisfied with food and service.

Zomato has observed that most couples order most of their food online. What is their average spending on each other?

```
df_online = df[df['online_order'] == 'Yes']
avg_spend_couples = df_online['approx_cost(for two people)'].mean()
print("Avg_spend_couples:", avg_spend_couples)
```

```
Avg_spend_couples: 510.3448275862069
```

- The Average spending by couples ordering food online is ₹510.34
- The Approx. Cost for two people is around ₹300 to ₹600 #From countplot of Approx cost for two people above
- This indicates a preference for moderately priced restaurants among couples using online ordering services.

Which mode (online or offline) has received the maximum rating?

```
mode_rating = df.groupby('online_order')['rate'].max()
mode_rating
```

```
online_order
No     4.3
Yes    4.6
Name: rate, dtype: float64
```

- Online orders have received higher ratings compared to offline orders
- This may be due to better convenience, discounts, ease of ordering, and improved customer experience in online platforms

Which type received more offline orders, so that Zomato can provide those customers with some good offers?

```
offline_type_counts = (df[df['online_order'] == 'No']
['listed_in(type)'].value_counts())

offline_type_counts
```

```
listed_in(type)
Dining    77
Cafes      8
Buffet     3
other      2
Name: count, dtype: int64
```

- Dining restaurants have received more offline orders compared to other restaurant types
- Zomato can target these customers with special offline dining offers and promotions to increase engagement and retention